

# INPUT

Tell the model how to read the values

01 / THE SAS EXAMPLE

```
data test;
  length char_num $8 char_date $9;
  input char_num $ char_date $;
  num_value = input(char_num, best12.);
  date_value = input(char_date, date9.);
  format date_value date9.;
  datalines;
123 15JAN2023
456 01FEB2023
789 31DEC2023
;
run;
proc print data=test noobs;
run;
```

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

**Make data types and date interpretation part of the request.**

Convert character values into numbers and dates. The nine-character date field preserves the full year. Check calendar dates and numeric values; Python need not copy SAS internal date storage.

EXPECTED RESULT

| char_num | char_date   | num_value | date_value |
|----------|-------------|-----------|------------|
| "123"    | "15JAN2023" | 123       | 2023-01-15 |
| "456"    | "01FEB2023" | 456       | 2023-02-01 |
| "789"    | "31DEC2023" | 789       | 2023-12-31 |

date\_value is displayed here as YYYY-MM-DD. SAS DATE9. displays the same calendar dates differently. Quotes mark source strings, not numeric outputs.

Expected values checked in Python; SAS execution pending.

# Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

## COPY THIS PROMPT + THE SAS EXAMPLE

```
Convert this SAS code to Python.  
Show code in this response. Do not create or run files, or read  
existing Python files.
```

## LOOK AT THE RESPONSE

Which library and method did it choose? Did it preserve the expected behavior? Compare the result with page 1 before accepting it.

## TRY IT / THREE FRESH ATTEMPTS

Run A three times with the same SAS example and setup. Save each first response before asking follow-up questions. Record agreement as well as differences.

Bedrock A/B trials for this example are pending.

## WHY CHOICES REMAIN OPEN

### 1 A recognizable task

The input values give useful clues. A short prompt may already produce valid numbers and dates.

### 2 Representation remains open

Numeric types, date objects, and display formats remain choices. Correct responses can differ in those choices.

### 3 A check is missing

Text can resemble a number or date while remaining unsuitable for the calculations that follow.

## ANTI-PATTERN TO AVOID

Anti-pattern: "Make the output look the same." Appearance cannot establish type or date meaning. Request numeric values and actual dates, then check both.

# Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

Convert this SAS code to Python using pandas.  
Show code in this response. Do not create or run files, or read existing Python files.

- Recreate char\_num and char\_date as strings, preserving row order.
- The three inputs are valid; month abbreviations are English.
- num\_value: use `pd.to_numeric(df['char_num'])`.
- date\_value: use `pd.to_datetime(df['char_date'], format='%d%b%Y')`. Keep this column as datetime values.
- Preserve both source columns and add the two result columns.
- Match calendar dates, not SAS's internal numeric date storage.
- Include assertions for num\_value = [123, 456, 789] and for date\_value formatted as YYYY-MM-DD = ['2023-01-15', '2023-02-01', '2023-12-31'].
- Also assert that the result columns have numeric and datetime types respectively. Calculate them; do not hard-code results.
- Return one complete script and a brief explanation of the format.

This exercise covers these valid inputs only.

WHY THESE DETAILS HELP

- 1

**Declare the starting point**  
Naming source columns as strings makes conversion explicit. Retaining them supports comparison.
- 2

**Explain the date pattern**  
The format specifies day, month abbreviation, and four-digit year, reducing choices about interpreting the text.
- 3

**Check meaning and type**  
Value checks confirm the dates and numbers. Type checks confirm they can be used as dates and numbers.

CHECK IT YOURSELF

Run the script separately. Compare all dates and numbers, then confirm column types. A displayed date string is not proof of a datetime column. These checks cover the supplied valid inputs.

**Specify what a value means and the type you need, then check both.**